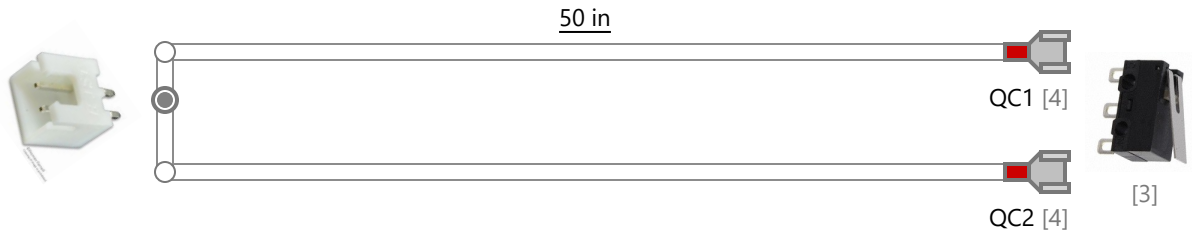


From	To	Conductor	Color	Size	From Contact PN
X axis Left Limit Switch.1	QC1	W1.Red		24 AWG	26-22 AWG Pin Crimp
X axis Left Limit Switch.2	QC2	W2.Black		24 AWG	26-22 AWG Pin Crimp

X axis Left Limit Switch [1, 2]

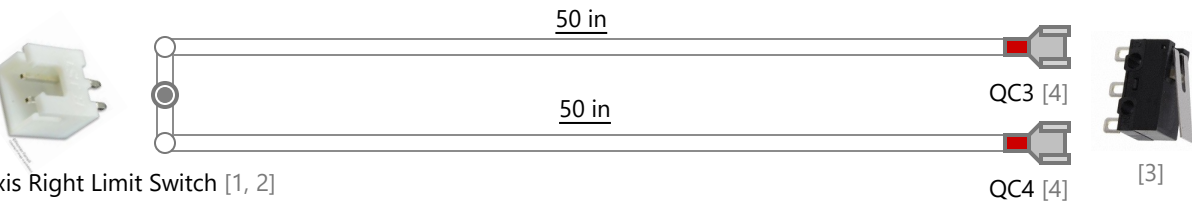


From	To	Conductor	Color	Size
QC1	X axis Left Limit Switch.1	W1.Red		24 AWG

From	To	Conductor	Color	Size
QC2	X axis Left Limit Switch.2	W2.Black		24 AWG

From	To	Conductor	Color	Size	From Contact PN
X axis Right Limit Switch.1	QC3	W3.Red		24 AWG	26-22 AWG Pin Crimp
X axis Right Limit Switch.2	QC4	W4.Black		24 AWG	26-22 AWG Pin Crimp

X axis Right Limit Switch [1, 2]

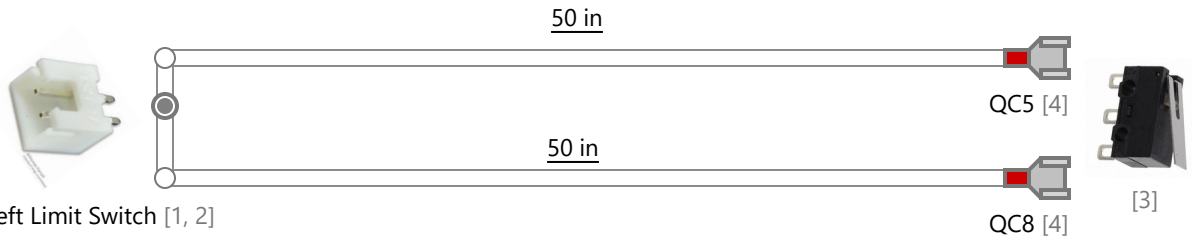


From	To	Conductor	Color	Size
QC3	X axis Right Limit Switch.1	W3.Red		24 AWG

From	To	Conductor	Color	Size
QC4	X axis Right Limit Switch.2	W4.Black		24 AWG

From	To	Conductor	Color	Size	From Contact PN
Y axis Left Limit Switch.1	QC5	W21.Red		24 AWG	26-22 AWG Pin Crimp
Y axis Left Limit Switch.2	QC8	W24.Black		24 AWG	26-22 AWG Pin Crimp

Y axis Left Limit Switch [1, 2]

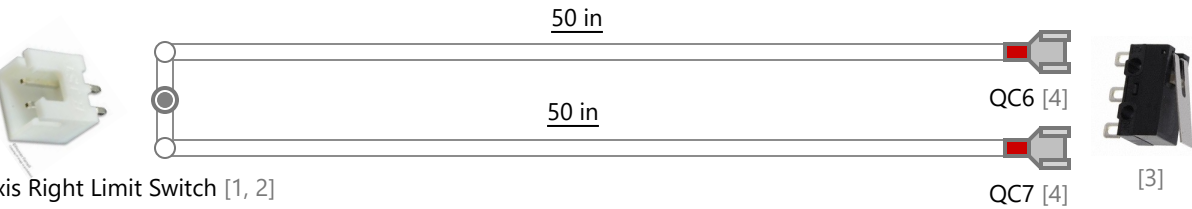


From	To	Conductor	Color	Size
QC5	Y axis Left Limit Switch.1	W21.Red		24 AWG

From	To	Conductor	Color	Size
QC8	Y axis Left Limit Switch.2	W24.Black		24 AWG

From	To	Conductor	Color	Size	From Contact PN
Y axis Right Limit Switch.1	QC6	W22.Red		24 AWG	26-22 AWG Pin Crimp
Y axis Right Limit Switch.2	QC7	W23.Black		24 AWG	26-22 AWG Pin Crimp

Y axis Right Limit Switch [1, 2]

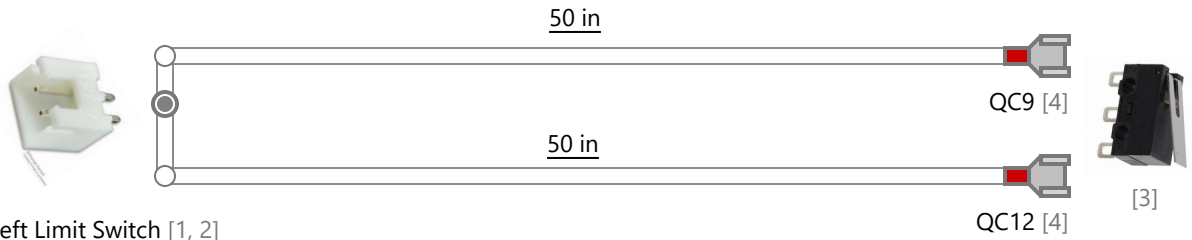


From	To	Conductor	Color	Size
QC6	Y axis Right Limit Switch.1	W22.Red		24 AWG

From	To	Conductor	Color	Size
QC7	Y axis Right Limit Switch.2	W23.Black		24 AWG

From	To	Conductor	Color	Size	From Contact PN
Z axis Left Limit Switch.1	QC9	W25.Red		24 AWG	26-22 AWG Pin Crimp
Z axis Left Limit Switch.2	QC12	W28.Black		24 AWG	26-22 AWG Pin Crimp

Z axis Left Limit Switch [1, 2]

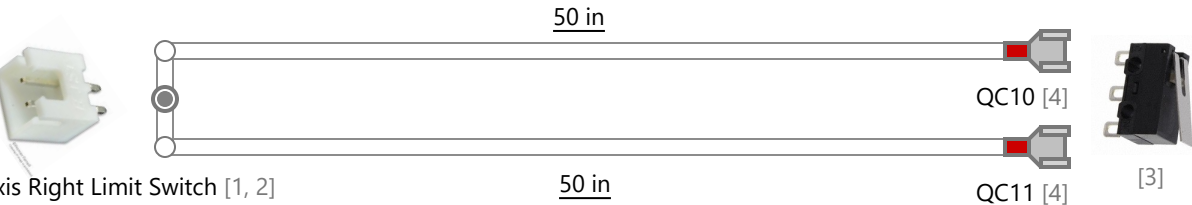


From	To	Conductor	Color	Size
QC9	Z axis Left Limit Switch.1	W25.Red		24 AWG

From	To	Conductor	Color	Size
QC12	Z axis Left Limit Switch.2	W28.Black		24 AWG

From	To	Conductor	Color	Size	From Contact PN
Z axis Right Limit Switch.1	QC10	W26.Red		24 AWG	26-22 AWG Pin Crimp
Z axis Right Limit Switch.2	QC11	W27.Black		24 AWG	26-22 AWG Pin Crimp

Z axis Right Limit Switch [1, 2]



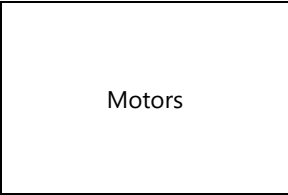
From	To	Conductor	Color	Size
QC10	Z axis Right Limit Switch.1	W26.Red		24 AWG

From	To	Conductor	Color	Size
QC11	Z axis Right Limit Switch.2	W27.Black		24 AWG

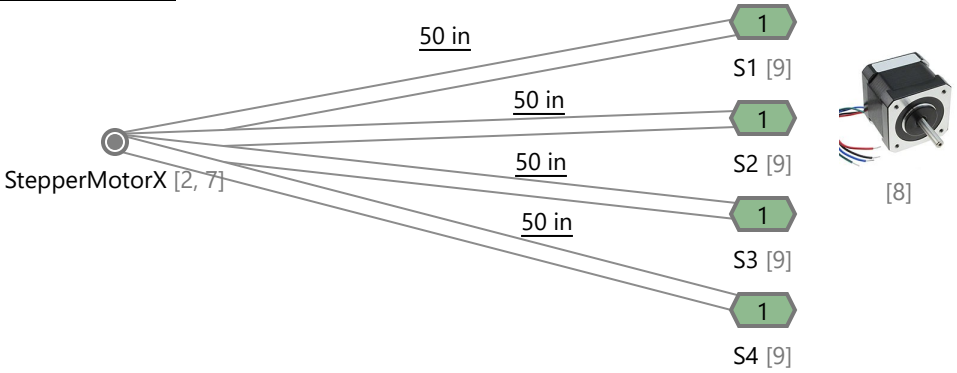
Bill of Materials

Id	Type	Manufacturer	Part Number	Quantity
1	Connector	JST	B2B-XH-A(LF)(SN)	7
2	Contact		26-22 AWG Pin Crimp	26
3			limit switch	6
4	Quick Connect Terminal		26 AWG Red Female	12
5	Wire		24 AWG Red	≥ 440 in
6	Wire		24 AWG Black	≥ 390 in
7	Connector	JST	B4B-XH-A(LF)(SN)	4
8			Stepper Motor	3
9	Splice		Western Union Splice	15
10	Wire		UL1015 24 AWG Green	≥ 50 in
11	Cable	Alpha Wire	6712 BL005	≥ 50 in
12			SG30 servo	1
13	Connector	Amphenol Communications Solutions	65039-036LF	6

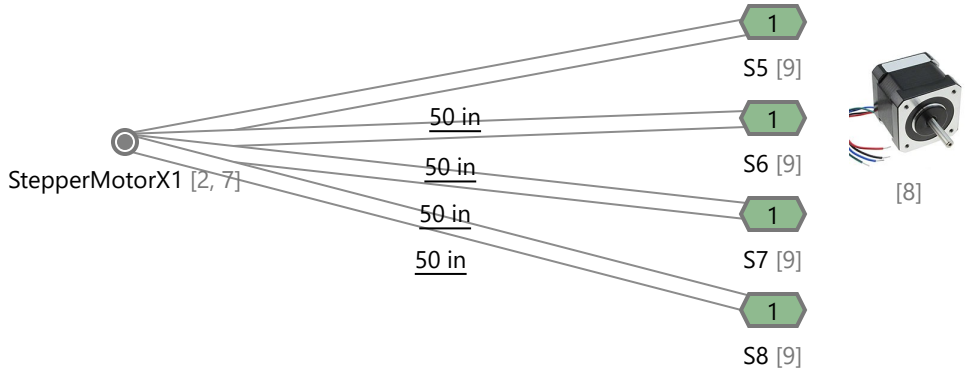
Limit Switches



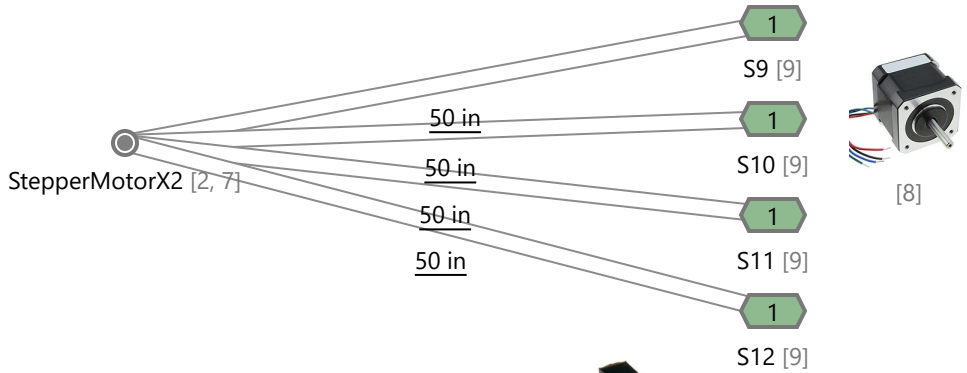
From	To	Conductor	Color	Size	From Contact PN
StepperMotorX.1	S1	W5.Red		24 AWG	26-22 AWG Pin Crimp
StepperMotorX.2	S2	W6.Black		24 AWG	26-22 AWG Pin Crimp
StepperMotorX.3	S3	W29.Green		24 AWG	26-22 AWG Pin Crimp
StepperMotorX.4	S4	CBL1.Unspecified		24 AWG	26-22 AWG Pin Crimp



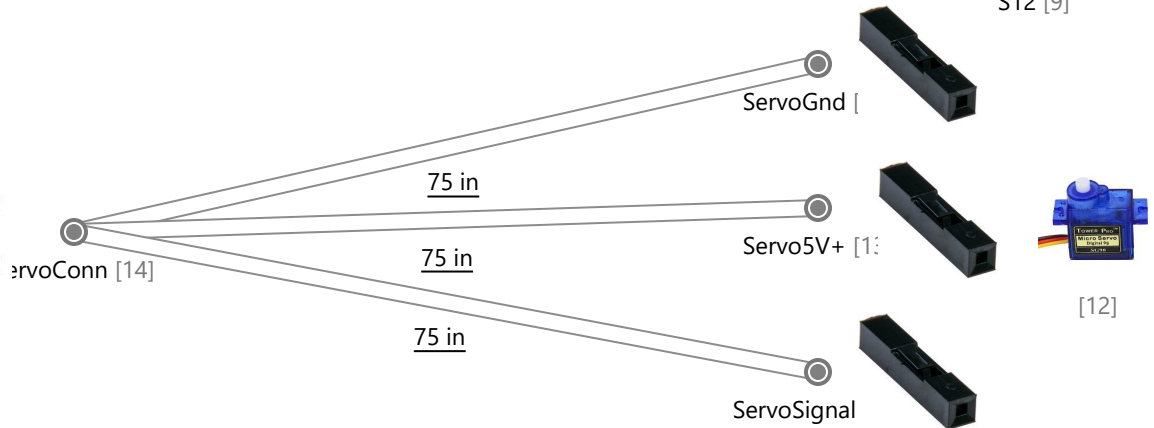
From	To	Conductor	Color	Size	From Contact PN
StepperMotorX1.1	S5	W30.Red		24 AWG	26-22 AWG Pin Crimp
StepperMotorX1.2	S6	W31.Black		24 AWG	26-22 AWG Pin Crimp
StepperMotorX1.3	S7	W32.Green		24 AWG	26-22 AWG Pin Crimp
StepperMotorX1.4	S8	CBL2.Unspecified		24 AWG	26-22 AWG Pin Crimp



From	To	Conductor	Color	Size	From Contact PN
StepperMotorX2.1	S9	W33.Red		24 AWG	26-22 AWG Pin Crimp
StepperMotorX2.2	S10	W34.Black		24 AWG	26-22 AWG Pin Crimp
StepperMotorX2.3	S11	W35.Green		24 AWG	26-22 AWG Pin Crimp
StepperMotorX2.4	S12	CBL3.Unspecified		24 AWG	26-22 AWG Pin Crimp



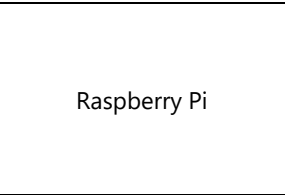
From	To	Conductor	Color	Size	From Contact PN
ServoConn.1	ServoGnd.1	W10.Black		24 AWG	
ServoConn.2	Servo5V+.1	W9.Red		24 AWG	
ServoConn.3	ServoSignal.1	W36.Yellow		24 AWG	



From	To	Conductor	Color	Size	From Contact PN
ServoGnd.1	ServoConn.1	W10.Black		24 AWG	

From	To	Conductor	Color	Size	From Contact PN
Servo5V+.1	ServoConn.2	W9.Red		24 AWG	

From	To	Conductor	Color	Size	From Contact PN
ServoSignal.1	ServoConn.3	W36.Yellow		24 AWG	



Raspberry Pi Communication [1, 2]

From	To	Conductor	Color	Size	From Contact PN
Raspberry Pi Communication.1	SCA.1	W7.Red	Red	24 AWG	26-22 AWG Pin Crimp
Raspberry Pi Communication.2	SCL.1	W8.Black	Black	24 AWG	26-22 AWG Pin Crimp



From	To	Conductor	Color	Size	From Contact PN
SCA.1	Raspberry Pi Communication.1	W7.Red	Red	24 AWG	

From	To	Conductor	Color	Size	From Contact PN
SCL.1	Raspberry Pi Communication.2	W8.Black	Black	24 AWG	



RaspPiPowerConn [7]

From	To	Conductor	Color	Size	From Contact PN
RaspPiPowerConn.1	RaspPiGnd.1	W10.Black	Black	24 AWG	
RaspPiPowerConn.2	RaspPiGnd1.1	W12.Black	Black	24 AWG	
RaspPiPowerConn.3	+5V.1	W13.Red	Red	24 AWG	
RaspPiPowerConn.4	+5V 1.1	W9.Red	Red	24 AWG	



From	To	Conductor	Color	Size	From Contact PN
RaspPiGnd.1	RaspPiPowerConn.1	W10.Black	Black	24 AWG	

From	To	Conductor	Color	Size	From Contact PN
RaspPiGnd1.1	RaspPiPowerConn.2	W12.Black	Black	24 AWG	

From	To	Conductor	Color	Size	From Contact PN
+5V.1	RaspPiPowerConn.3	W13.Red	Red	24 AWG	

From	To	Conductor	Color	Size	From Contact PN
+5V 1.1	RaspPiPowerConn.4	W9.Red	Red	24 AWG	